

FieldCommand Incident Management System

Open-Source · Offline-First · Field-Deployable

FIELDCOMMAND

Incident Management System v1.0

FIELD QUICK REFERENCE

Operator Quick Reference Card

RACES · ARES · ICS/NIMS · Amateur Radio EMCOMM · Public Safety

July 2026

■ Getting Started & Dashboard

<http://192.168.50.1>

STEP	WHAT TO DO
1	Connect to EMCOMM-NET — open Wi-Fi on your device and select EMCOMM-NET.
2	Open any browser — type http://192.168.50.1 and press Enter.
3	Select a mode — three tabs at the top: Amateur Radio · Public Safety · All-Hazards ICS.
4	Bookmark it — on a phone tap Share → Add to Home Screen for one-tap access.

MODE	USE IT FOR
All-Hazards ICS	Incident management, IAP, T-cards, FEMA costs, personnel check-in
Amateur Radio EMCOMM	ARES/RACES nets, APRS, Winlink, JS8Call, AMPRNet, callsign lookup
Public Safety	Starcom/P25 nets, resource map, weather radar, hospital directory

INDICATOR	MEANING
■ Green — Cellular	Primary internet active — all live features enabled
■ Blue — Satellite	Satellite failover active — all live features enabled
■ Red — Offline	No internet — ALL core ICS tools still fully operational
■ Green — AMPRNet UP	44Net WireGuard tunnel active — gateway Pi reachable — 44.x.x.x address shown
■ Red — AMPRNet DOWN	44Net tunnel down or gateway Pi unreachable — amateur radio group only

■ A red WAN indicator does NOT mean FieldCommand is down. Net loggers, ICS forms, T-cards, roster, check-in, and all local tools run offline.

■ Amateur Net Control Logger

<http://192.168.50.1/netcontrol.html>

STEP	WHAT TO DO
1	Open a net — enter net name, frequency, and mode. Click Open Net .
2	Check in a station — type callsign and press Enter. Name auto-fills from offline FCC database.
3	Log traffic — click + Traffic — enter precedence, from, to, and message info.
4	Check out a station — click Check Out . Duration recorded to nearest ¼ hour.
5	Close the net — click Close Net . All remaining stations auto-checked out.
6	Export ICS-309 — downloads formatted comms log with all check-ins and durations.
7	Observer link — click ■ to share a read-only live view with EOC or served agency.

■ Public Safety Net Logger

<http://192.168.50.1/starcom.html>

FIELD	DESCRIPTION
Radio ID	Trunked/P25 unit number — required for every check-in
Unit Name	Agency and unit description. Roster auto-fills if Radio ID matches.
Dispatch Ctr	Dispatching agency (e.g. MCECC, IDOT)
Export ICS-309	Radio IDs appear in station column — same export as amateur logger

■ Member Roster & QR Check-In Codes

<http://192.168.50.1/roster.html>

ACTION	HOW
Add member	Click + Add Member — enter Member ID, name, callsign or Radio ID, role
Import CSV	Click ■ Import CSV — required column: member_id. All others optional.
Generate QR code	Click QR button on any member card — print or save PNG for their badge
Export roster	Click ■ Export CSV — downloads all members for backup or transfer

■ Scan Check-In (QR/Barcode)

http://192.168.50.1/scan_checkin.html

STEP	WHAT TO DO
1	Open Scan Check-In on a camera device. Click ■ Start Camera .
2	Scan QR code — point camera at member's QR. Name auto-fills from roster.
3	Manual fallback — type member ID, callsign, or Radio ID and press Enter.
4	Check In — confirm fields and click ✓ Check In. Success screen appears.

■ Personnel Accountability (PAR)

<http://192.168.50.1/accountability.html>

TAB	WHAT IT SHOWS
ICS-211 Check-In List	All checked-in personnel · Check Out button · PAR confirm button
T-Card Personnel	Personnel grouped by resource · PAR per resource group
■ Unaccounted	Personnel not yet confirmed in current PAR — contact supervisor immediately
Cross-Reference	Checked-in but no T-card · On T-card but no check-in (safety gap)

■ Click ■ Conduct PAR to start a PAR cycle. Click ✓ PAR on each person as supervisors report in. Reset PAR to begin a new cycle.

■ T-Card Resource Board & Personnel

<http://192.168.50.1/resources.html>

STEP	WHAT TO DO
1	Add resource — click + Add Resource. Enter name, type, status, assignment, leader.
2	Change status — click status badge to cycle: Available → Assigned → Staging → Out of Service.
3	Add personnel — click any T-card → ■ PERSONNEL tab → type name → select from roster suggestion.
4	Set ICS position — choose the person's role on THIS resource (Crew Boss, EMT, Operator, etc.).

STATUS	COLOR	MEANING
Available	Green	On scene and ready for assignment
Assigned	Blue	Actively tasked to a division or group
Staging	Lt Blue	En route or waiting at staging area
Out of Service	Red	Unavailable — mechanical, medical, or released

■ GPS-Tracked Resource Map

http://192.168.50.1/resource_map.html

METHOD	HOW TO SET POSITION
Device GPS	Click ■ Use Device GPS — browser gets your GPS coordinates automatically
Click map	Click ■ Click Map to Place — crosshair cursor, click the map to set position
Manual coords	Type decimal latitude and longitude directly into the fields

■ Resource pins are color-coded by status. Resources without GPS show as dashed circles in the sidebar. Enable Auto-refresh 30s for moving resources.

LAYER	HOW TO ENABLE
■ Repeater overlay	Click ■ Repeaters in the toolbar — loads all repeaters from the local database and plots mode-colored pins on the resource map. Amber pins = ARES/RACES-affiliated. Click any pin for frequency, offset, and tone.

■ ICS Platform — Five-Section Structure

<http://192.168.50.1/ics/>

STEP	WHAT TO DO
1	Create incident — go to http://192.168.50.1/incident.html → + New Incident.
2	Navigate sections — tabs at top: ■ Command · ■ Operations · ■ Planning · ■ Logistics · ■ Finance.
3	Advance period — click + New Period at start of each operational period.
4	Use event template — event_templates.html → Activate → select type → instant pre-built incident.

SECTION	KEY TOOLS & FORMS
■ Command	ICS-201 briefing · ICS-202 objectives · ICS-203 org chart · ICS-208 safety message · Position checklists · Meeting scheduler
■ Operations	T-card resource board · ICS-204 assignment lists · GPS resource map · Resource personnel rosters
■ Planning	IAP assembly · ICS-205 comms plan · ICS-206 medical plan · ICS-209 status summary · ICS-211 check-in list · Planning P cycle
■ Logistics	ICS-205 radio comms (COML develops) · ICS-205A comms list · ICS-206 medical plan (MEDL develops) · Channel library · Facilities · ICS-218
■ Finance/Admin	FEMA PA cost tracking · Force Account Labor · Equipment · Materials · Contracts · Cost dashboard · ICS-214 activity log

■ Under Unified Command (UC): enter "Unified Command" in the IC field. List UC members in the ICS-203 Organization Assignment List. All forms and T-cards work identically under UC.

■ ICS Forms, Digital Signatures & IAP Compilation

<http://192.168.50.1/ics-form.html>

FORM	TITLE	DEVELOPED BY
ICS-202	Incident Objectives	Planning Section Chief / IC
ICS-203	Org Assignment List	Resources Unit Leader (RESL)
ICS-204	Assignment List	RESL — distributed to Operations
ICS-205	Radio Comms Plan	Comms Unit Leader (COML) — Logistics
ICS-206	Medical Plan	Medical Unit Leader (MEDL) — Logistics
ICS-207	Org Chart	Resources Unit Leader (RESL)
ICS-208	Safety Message	Safety Officer (SOFR) — Command Staff
ICS-209	Status Summary	Situation Unit Leader (SITL)
ICS-211	Check-In List	Resources Unit Leader (RESL)
ICS-213	General Message	Any ICS position
ICS-214	Activity Log	All supervisors — every section
ICS-215A	IAP Safety Analysis	Safety Officer (SOFR) with Ops input
ICS-218	Support Vehicle Inventory	Ground Support Unit Leader (GSUL)
ICS-221	Demobilization Check-Out	Demob Unit Leader (DMOB)
ICS-309	Communications Log	Comms Unit Leader (COML) / any operator

TASK	HOW
Digital signature	Click any Prepared By / Approved By field — sign with mouse, touch, or stylus → Accept
Print IAP on-site	iap.html → select forms + period → ■ Print IAP → browser print dialog opens.
Save IAP as PDF (recommended)	iap.html → select forms + period → ■ Save as PDF. Pi generates proper 8.5×11 PDF → downloads as .pdf. Copy to USB or email — opens in any PDF viewer.
Save IAP as HTML (fallback)	iap.html → select forms + period → ■ Save as HTML. No server needed — browser generates HTML file directly. Open in any browser → File → Print. Page layout may vary by browser.
Full PDF compiler	iap_compile.html → select forms → Download IAP PDF (5–15 sec). Server-side PDF with cover page, section dividers, embedded signatures.

■ FEMA PA Cost Documentation

http://192.168.50.1/fema_costs.html

CATEGORY	WHAT TO ENTER	FEMA REQUIREMENT
Force Account Labor	Employee name, title, regular hours, overtime hours, hourly rate, fringe %	Only overtime is reimbursable for employees; all hours for volunteers
Equipment	Click ■ Lookup → select FEMA category → rate auto-fills → enter hours and unit ID	Must use FEMA Schedule of Equipment Rates (44 categories pre-loaded)
Materials	Vendor, description, PO number, unit cost, quantity	Source documentation required — attach receipts to Project Worksheet
Contracts	Vendor, description, contract number, total amount	Contract documentation required — competitive bid or sole-source justification

STEP	WHAT TO DO
1	Update FEMA rates — go to fema_rates.html annually when FEMA publishes new schedule.
2	Import ICS-214 — on Force Account Labor tab, click Import from ICS-214 to pull hours.
3	Export PW text — click ■ Export PW Text for a formatted Project Worksheet summary.

■ Real-Time Cost Dashboard

http://192.168.50.1/cost_dashboard.html

PANEL	WHAT IT SHOWS
Total incident cost	Running total — updates every 2 minutes automatically
Cost by category	Labor · Equipment · Materials · Contracts — amount and % of total
Cost projection	Extrapolated cost based on current burn rate × remaining periods
Budget tracker	Enter authorized budget to see remaining balance and % used
Per-period breakdown	Costs attributed by operational period — for AAR documentation

■ NEXRAD Animated Radar

<http://192.168.50.1/radar.html>

CONTROL	FUNCTION
■ / ■ Play/Pause	Start or stop animation loop
■ ■ Step	Advance one frame at a time
Scrubber	Jump to any point in the loop
Speed	0.5× · 1× · 2× · 4× playback speed
Station	Select nearest NEXRAD radar station for your area
Palette	Standard (reflectivity) · Dual-pol · Enhanced
Auto-refresh	New tiles load every 5 minutes when WAN is active

■ WAN REQUIRED for radar tiles. When offline, last loaded frame displays with timestamp. Radar resumes automatically when WAN returns.

■ Tactical APRS Map

<http://192.168.50.1/tactical.html>

SOURCE	REQUIRES WAN?	NOTES
RF APRS (TNC/direwolf)	✗ Always offline	Live RF stations from local TNC — always available
APRS-IS internet feed	✓ WAN required	Internet feed pauses when offline; RF continues
SARTopo GeoJSON overlay	✗ Once imported	Import at sartopo_import.html — persists on map
NWS weather polygons	✓ WAN required	Alert polygons overlay on the map

■ Amateur Radio EMCOMM Tools

<http://192.168.50.1/>

■ **AMPRNet / 44Net Gateway applies only when an amateur radio group leads or is a key partner in the deployment.** FCC amateur license required for registration — a public safety agency or served organization cannot register independently. All other tools on this page work for any deployment. See User Manual Ch. 28 and Installation Guide Step 11 for full details.

TOOL	URL	QUICK USE
FCC Callsign Lookup	/callsign.html	Type callsign → Enter. 800K+ licensees, fully offline. Click + Add to Roster.
NTS Radiogram	/nts.html	Fill preamble fields — word count auto-calculates. Print for traffic handling.
Winlink Import	/winlink-import.html	Drag Winlink XML form attachment → Review → Archive to active incident.
JS8Call	Windows laptop	Click JS8Call dashboard card → enter laptop IP → opens JS8Call web interface.
HF Propagation	/propagation.html	SFI · K-index · A-index from NOAA SWPC. Band-by-band conditions. WAN required.
AMPRNet Gateway	192.168.50.2	44Net WireGuard tunnel status · callsign-authenticated · Part 97 access log.
Observer Mode	/observer.html	Read-only live net view. Share URL with EOC — auto-refreshes every 15 seconds.
Dead Man's Switch	/deadmans.html	Arm next to a net. Audible alert if net goes silent beyond configured interval.
ICS-309 Manual	/ics309.html	Manual comms log entries. Add timestamp · from · to · subject · precedence.
ICS-213 Message	/ics213.html	Formal written message with digital signature. Print or send via Winlink.

■ Reference Tools

<http://192.168.50.1/>

TOOL	URL	QUICK USE
Repeater Database	/repeaters.html	Filter by band/mode/CTCSS. Add local repeaters. Feeds ICS-205 channel picker.
Channel Library	/channel_library.html	Pre-loaded interop channels (NIFOG/VTAC). Add custom agency channels.
Grid Square Calc	/grid.html	Grid ↔ lat/lon · distance and bearing between two grids.
Hospital Directory	/hospitals.html	Trauma level · helipad · phone. Feeds ICS-206 medical plan.
Facilities Directory	/facilities.html	ICP · staging areas · shelters · supply depots. Links to T-card assignments.
NIMS Resource Types	/resource_types.html	Pre-loaded NIMS types. Add custom types — available instantly in T-card board.
Radio Cheat Sheets	/cheatsheets.html	Phonetic alphabet · Q-codes · RST scale · ICS structure · HF calling frequencies.
Position Checklists	/position_checklists.html	NIMS activation checklists for IC, SOFR, OSC, PSC, LSC, FSC and unit leaders.
Reference Library	/refs.html	Kiwix offline Wikipedia, WikiMed, iFixit. No internet needed.
Pre-Flight Check	/preflight.html	GO/CAUTION/NO-GO readiness verdict. Run before every activation.
Event Templates	/event_templates.html	6 built-in templates (Shelter/SAR/HazMat/Weather/Mass Gathering/Exercise). Activate → incident pre-loaded with objectives, resources, channels, org.
Incident Archive	/incident_mgmt.html	Archive to USB · Restore from USB · Hard delete · Beta/Scenario Reset.

■ WAN Configuration & Network Reference

http://192.168.50.1/wan_settings.html

SETTING	OPTIONS
Role	Preferred (used first) · Fallback (used when preferred fails)
Type	Cellular · Satellite · Hotspot · Fixed broadband · Other
Detection	internet_only (hotspots/dongles) · ping (gateway IP) · admin_reachable (modem UI)
Swap roles	Click ■ Swap Roles — preferred and fallback swap immediately
Enable/disable	Toggle a source on or off without deleting its configuration

■ Admin URLs & IP Reference

<http://192.168.50.1/>

DEVICE / SERVICE	ADDRESS	NOTES
FieldCommand dashboard	http://192.168.50.1	Main entry — all tools
WAN status detail	/wan-status.html	Both sources · last check · event log
ICS platform	http://192.168.50.1/ics/	Five-section ICS structure
AMPRNet gateway status	http://192.168.50.2	Read-only port 9000
Primary router admin	http://192.168.50.254	ASUS RT-BE58 Go · AiMesh · WAN config
CUPS printer admin	http://192.168.50.1:631	Add/manage USB printers
Health monitor	http://192.168.50.1:5051/health	CPU · disk · service dots · GPS
FCC lookup API	http://192.168.50.1:5050	Port 5050 — callsign, roster, WAN config
ICS platform API	http://192.168.50.1:5051	Port 5051 — forms, T-cards, FEMA, GPS
Kiwix library	http://192.168.50.1:8080	Offline Wikipedia and reference docs
AMPRNet gateway	http://192.168.50.2	44Net status · callsign auth · Part 97 log
Planning P cycle	/ics/planningp.html	15-phase IAP cycle guide with agenda and required forms
Starcom resource map	/resmap.html	Public safety unit positions — separate from ICS resource map
ICS-214 Activity Log	/ics214.html	Unit log — export hours directly to FEMA Force Account Labor
Meeting scheduler	/meetings.html	Schedule operational briefings — time, location, attendees
Org setup	/setup.html	Configure org name, callsign, grid square, time zone
ICS-204A Briefing Sheet	/briefing_204a.html	Assignment briefing sheet for branch/division supervisors

MESH NODE	SWITCH PORT	PLACEMENT
FieldCommand Pi 5	Port 1	Primary server — rack or case shelf
44Net Gateway Pi 5	Port 2	AMPRNet gateway — rack or case shelf
ASUS RT-BE58 Go (primary)	Port 3	ICP primary Wi-Fi · AiMesh controller
RT-BE58 Go (node 1)	Port 4	Coverage extension — second room or floor

MESH NODE	SWITCH PORT	PLACEMENT
RT-BE58 Go (node 2)	Port 5	Coverage extension — outdoor or staging

■ Troubleshooting

<http://192.168.50.1/>

SYMPTOM	LIKELY CAUSE	FIX
Cannot reach 192.168.50.1	Not on EMCOMM-NET Wi-Fi	Check Wi-Fi — must show EMCOMM-NET. Some devices auto-switch networks.
Page loads but tools error	API server not running	SSH to Pi: <code>sudo systemctl restart fieldcommand-main fieldcommand-fcc</code>
Callsign lookup fails	FCC database missing or callsign not US licensed	Run: <code>sudo python3 /opt/fieldcommand/scripts/build_fcc_db.py</code>
APRS map shows no stations	No RF or APRS-IS unavailable	Check TNC/direwolf connection. WAN needed for APRS-IS feed.
Radar shows offline	No internet connection	Red WAN indicator means no radar tiles. All other tools still work.
QR scan not working	Browser does not support BarcodeDetector	Use Chrome or Edge. Use manual ID entry on Firefox/Safari.
IAP PDF takes too long	Many forms or large signatures	Normal 5–20 sec. If over 60 sec check server health at <code>:5051/health</code> .
WAN shows red but internet works	Detection method misconfigured	Go to <code>wan_settings.html</code> . Change detection to <code>internet_only</code> .
T-card personnel not saving	No active incident selected	Confirm incident is active at <code>incident.html</code> before adding personnel.
Gateway Pi unreachable	Gateway Pi off or not on switch	Check gateway Pi power (green LED). Confirm connected to UniFi switch.
PAR shows wrong count	Personnel checked in without ICS-211	Run Cross-Reference tab on <code>accountability.html</code> to find gaps.



For any issue not covered here — open the Health Monitor at <http://192.168.50.1:5051/health> or check the service log: `journalctl -u fieldcommand-main -n 50`